



LOVELY
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School of Computer Science and Engineering

SUMMER TRAINING/INTERNSHIP

PROJECT REPORT

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Loan Approval Prediction Using Machine Learning

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Acknowledgement

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Abstract

Loan approval decisions made manually by lending institutions are often slow, inconsistent, and susceptible to reviewer bias. This project presents an end-to-end data science solution for predicting loan approval outcomes using machine learning, built on a dataset of 20,000 loan applicant records sourced from Kaggle.

The project workflow began with data cleaning and feature engineering, followed by an extensive Exploratory Data Analysis (EDA) phase that used correlation analysis, distribution plots, and outlier detection to understand the dataset. Statistical hypothesis testing, including independent T-tests, a Chi-Squared test, a Shapiro-Wilk normality test, and a Variance Inflation Factor (VIF) multicollinearity check, was used to validate observed patterns before they were used for feature selection.

Three candidate classification models were trained and compared: Logistic Regression, Decision Tree, and Random Forest. During model development, a critical data leakage issue was identified and corrected: the RiskScore column was found to be almost perfectly correlated with the target variable and was excluded from the feature set to avoid producing a misleadingly high accuracy. After this correction, Logistic Regression emerged as the best-performing model, achieving 95.7% accuracy, 91.7% precision, 90.3% recall, and an F1-score of 91.0% on unseen test data.

Finally, all analytical findings and model outputs were synthesised into a multi-page, interactive Power BI dashboard, providing lending professionals with an intuitive, filterable view of applicant trends, risk factors, and model performance. The complete project, including source code and dashboard file, has been published on GitHub and shared on LinkedIn.

Keywords: Loan Approval Prediction, Machine Learning, Logistic Regression, Exploratory Data Analysis, Data Leakage, Power BI, Business Intelligence.

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Chapter 1: Introduction

1.1 Institution / Company Profile

This project was undertaken as a capstone initiative under the auspices of [Lovely Professional University]. The Department of Computer Science is dedicated to fostering academic excellence and practical innovation, providing students with the resources and guidance to solve complex, real-world problems.

The internship simulates a real-world business scenario, treating a financial lending institution as the client organization. The objective was to apply the theoretical concepts of machine learning, statistical analysis, and data visualization learned throughout the curriculum to a comprehensive dataset, mimicking the workflow and deliverables expected in a professional data analytics role.

Throughout the internship, emphasis was placed not only on producing a working predictive model, but on following the complete, disciplined data science lifecycle — from raw data acquisition through to a polished, decision-ready business intelligence dashboard — in a manner consistent with how such projects are executed in industry.

1.2 Overview of Training Domain

The project was conducted in the domain of Financial Data Analytics, focusing on leveraging data science techniques to solve a critical problem faced by lending institutions: assessing the risk of loan applicants in an objective, data-driven manner. The work involved handling a large applicant dataset, applying statistical methods, building predictive classification models, and creating comprehensive dashboards for data visualization and business intelligence.

Financial institutions traditionally rely on manual underwriting processes that are time-consuming, inconsistent between reviewers, and prone to unconscious bias. Data-driven approaches, such as the one developed in this project, offer a scalable and objective alternative that can be audited, monitored, and continuously improved.

1.3 Objective of the Project

The primary objective was to develop an end-to-end solution for predicting and visualizing loan approval outcomes. This involved performing Exploratory Data Analysis (EDA) to uncover applicant trends and risk factors, building and comparing multiple classification models (Logistic Regression, Decision Tree, and Random Forest) to predict loan approval, and developing an interactive multi-page Power BI dashboard to present these insights in an actionable format for lending professionals.

The specific sub-objectives of the project were to:

- Understand and document the structure, quality, and characteristics of the loan applicant dataset.
- Apply rigorous statistical hypothesis testing to validate observed patterns before acting on them.
- Build multiple candidate machine learning models and select the best one using appropriate evaluation metrics rather than accuracy alone.
- Identify and correct data quality issues, including data leakage, that could otherwise produce misleading results.
- Translate all findings into a business-ready Power BI dashboard suitable for non-technical stakeholders.

1.4 Project Links

The complete source code, notebooks, and Power BI file for this project are hosted publicly and can be accessed using the links below:

GitHub Repository:

<https://github.com/utkarshsingh287>

LinkedIn Post:

<https://www.linkedin.com/in/utkarsh-raj-singh-702a08334/>

Chapter 2: Training Overview

2.1 Tools & Technologies Used

Programming & ML: Python, Pandas, NumPy, Scikit-learn, SciPy, Statsmodels

Data Visualization: Matplotlib, Seaborn, Microsoft Power BI, Power Query

Development Environment: Jupyter Notebook

Version Control & Sharing: Git, GitHub, LinkedIn

2.2 Areas Covered During Training

- Data Cleaning and Preprocessing
- NumPy array operations and Pandas Series/DataFrame manipulation
- Statistical Analysis and Hypothesis Testing (T-test, Chi-Squared, Shapiro-Wilk, VIF, A/B Testing)
- Probability Distributions (Normal, Uniform, Binomial, Poisson)
- Machine Learning: Logistic Regression, Decision Tree, and Random Forest for Binary Classification
- Model Evaluation and Interpretation using Accuracy, Precision, Recall, F1-Score, ROC-AUC
- Business Intelligence and Dashboard Design with Power BI
- Publishing and documenting a project on GitHub and sharing insights on LinkedIn

2.3 Summary of Work

The project followed a structured, sequential workflow to ensure a thorough and logical analysis. It commenced with data acquisition and rigorous cleaning of a 20,000-row applicant dataset, involving duplicate checks, missing value checks, and the removal of non-predictive fields. This was followed by an extensive Exploratory Data Analysis (EDA) phase, which utilised statistical tests and visualisations — including histograms, correlation heatmaps, boxplots, and hypothesis testing — to identify key predictors and underlying patterns in the data.

Next, three machine learning models were developed and evaluated; a Logistic Regression algorithm was found to perform best, achieving 95.7% accuracy, and its performance was assessed using precision, recall, F1-score, and a confusion matrix. Finally, all insights and

results were synthesised into a multi-page, interactive Power BI dashboard, designed to visualise applicant trends, demographics, and risk factors dynamically for end-users.

This workflow closely mirrors the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, which is discussed in more detail in Chapter 6, and reflects standard practice in professional data analytics roles.

Chapter 3: Project Details

3.1 Title of the Project

Loan Approval Prediction Using Machine Learning

3.2 Problem Definition

Manual and subjective assessment of loan applicant risk leads to inefficiencies, inconsistency, and potential oversights for lending institutions. This project addresses the need for an automated, data-driven system to objectively identify applicants likely to be approved or rejected for a loan, facilitating faster, fairer, and more consistent lending decisions.

3.3 Scope and Objectives

Scope: The project covers the analysis of a historical loan applicant dataset, development of a predictive classification model, and creation of a visual dashboard prototype.

Objectives:

- To preprocess applicant data and check for missing values, duplicates, and inconsistencies.
- To perform EDA and statistical testing to identify key factors influencing loan approval.
- To build, train, and evaluate multiple classification models for binary approval prediction.
- To design a multi-page Power BI dashboard to visualise all insights and model results.
- To publish the project on GitHub and share key insights publicly on LinkedIn.

3.4 System Requirements

The system must process a CSV dataset, clean it, build a predictive model with strong classification performance, and provide an interactive dashboard with filters and key metrics.

Software: Python 3.10+, Jupyter Notebook, Power BI Desktop

Python Libraries: pandas, numpy, matplotlib, seaborn, scikit-learn, scipy, statsmodels

Hardware: A standard laptop/desktop with at least 8 GB RAM is sufficient for this dataset size (20,000 rows).

3.5 Data Source Description

The “Loan Approval” dataset, sourced from Kaggle, was used. This dataset contains 20,000 synthetic loan applicant records, including demographics, income, credit history, assets, liabilities, and loan details, ideal for building predictive models without privacy concerns.

Source: https://www.kaggle.com/datasets/lorenzozoppelletto/financial-risk-for-loan-approval?utm_source=chatgpt.com

Key columns used in this project include: Age, AnnualIncome, CreditScore, EmploymentStatus, EducationLevel, MaritalStatus, HomeOwnershipStatus, LoanAmount, LoanDuration, LoanPurpose, DebtToIncomeRatio, TotalAssets, TotalLiabilities, NetWorth, and the target variable LoanApproved.

Chapter 4: Implementation

4.1 Tools and Technologies

As listed in Chapter 2.1. All development was carried out in Python using Jupyter Notebook, with the final business-facing deliverable built in Microsoft Power BI.

4.2 Methodology

A straightforward, project-focused methodology was employed: Data Preparation, Exploration, Statistical Testing, Modelling, and Visualisation, in line with the CRISP-DM framework.

4.3 Modules and Implementation Details

4.3.1 Data Preprocessing and Feature Engineering

- The dataset was checked for missing values and duplicate rows; none were found, confirming the dataset was pre-cleaned.
- The ApplicationDate column was removed as it did not contribute predictive value.
- New features were engineered from existing columns to improve model performance, including LoanToIncomeRatio, LiquidAssets, DebtToAssetRatio, and AgeGroup.
- Categorical columns (EmploymentStatus, EducationLevel, MaritalStatus, HomeOwnershipStatus, LoanPurpose) were label-encoded for use in the models.
- All numeric features were scaled using StandardScaler so that no single feature dominated the model due to its magnitude.

Code for Feature Engineering:

```
df['LoanToIncomeRatio'] = df['LoanAmount'] / (df['AnnualIncome'] + 1)
df['LiquidAssets'] = df['SavingsAccountBalance'] + df['CheckingAccountBalance']
df['DebtToAssetRatio'] = df['TotalLiabilities'] / (df['TotalAssets'] + 1)
df['AgeGroup'] = pd.cut(df['Age'], bins=[18, 30, 45, 60, 100],
                        labels=['Young', 'Adult', 'MiddleAge', 'Senior'])
```

4.3.2 Exploratory Data Analysis (EDA) and Statistical Testing

- Histograms and boxplots were generated for all key numerical features to understand their distribution and identify outliers.
- A correlation heatmap was created to identify relationships between features.
- Scatter plots, such as Annual Income vs Loan Amount, were used for bivariate analysis.
- T-tests and a Chi-Squared test were conducted to confirm the statistical significance of differences in key indicators between approved and rejected applicant groups.
- A Shapiro-Wilk test was used to check the normality of Annual Income, and a Variance Inflation Factor (VIF) check was used to confirm low multicollinearity among financial features.

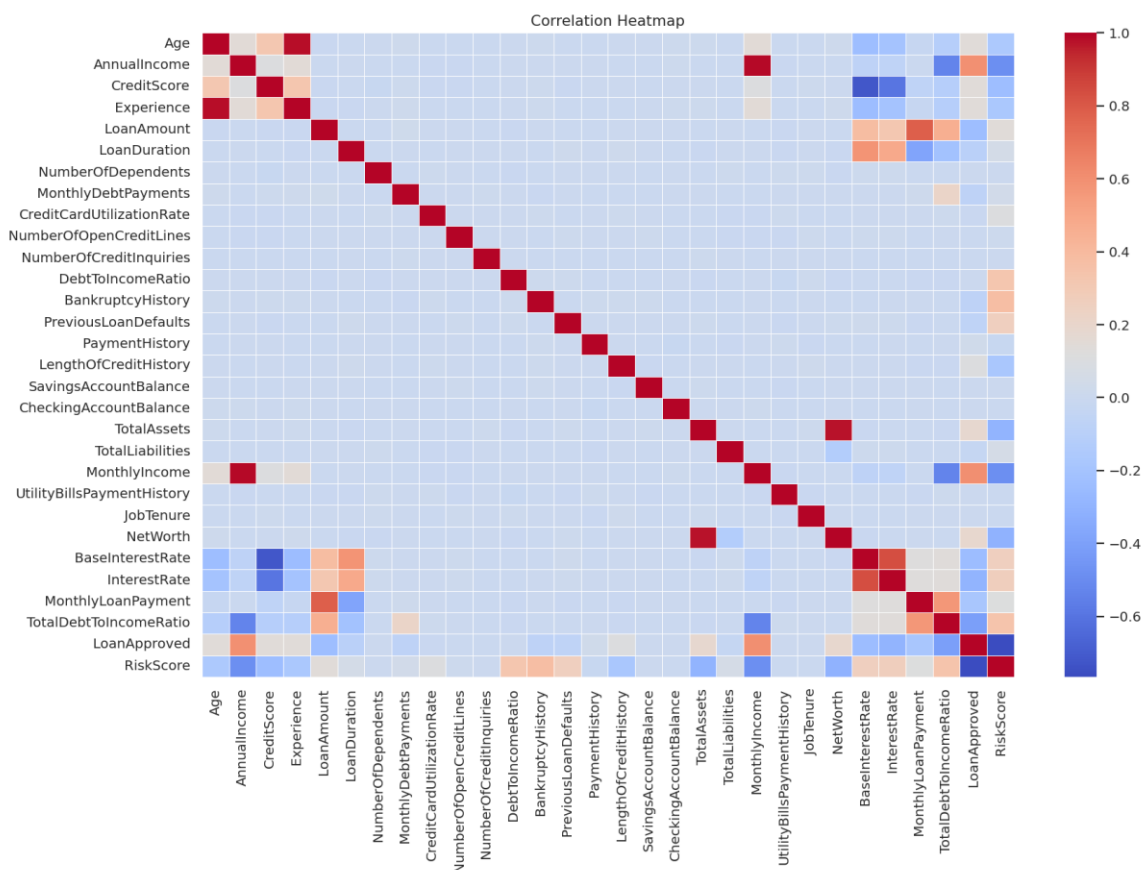


Figure 1: Correlation Heatmap of all numeric features

Code for Heatmap:

```
plt.figure(figsize=(16, 10))
correlation = df.corr(numeric_only=True)
sns.heatmap(correlation, cmap='coolwarm', annot=False, linewidths=0.5)
plt.title('Correlation Heatmap of All Numeric Features')
```

```
plt.tight_layout()  
plt.show()
```

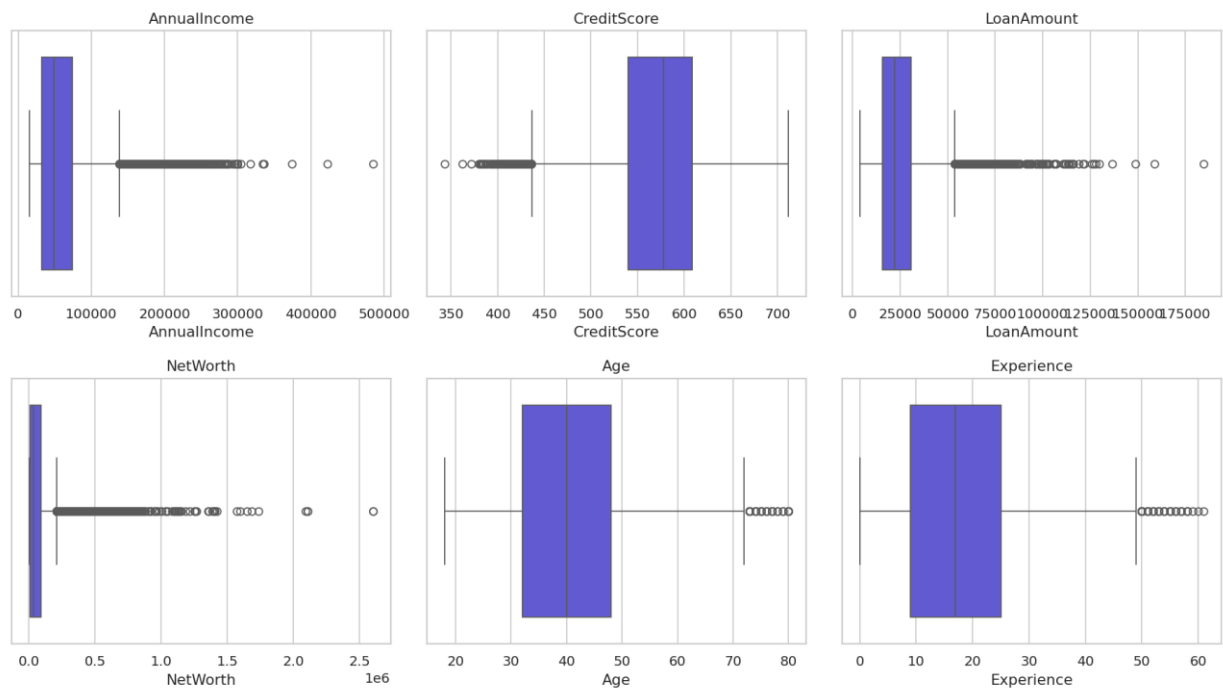


Figure 2: Boxplots of key numerical features (outlier detection)

Code for Boxplots:

```
outlier_cols = ['AnnualIncome', 'CreditScore']
for col in outlier_cols:
    plt.figure()
    sns.boxplot(x=df[col])
    plt.title(f'{col} - Outlier Detection')
    plt.tight_layout()
    plt.show()
```

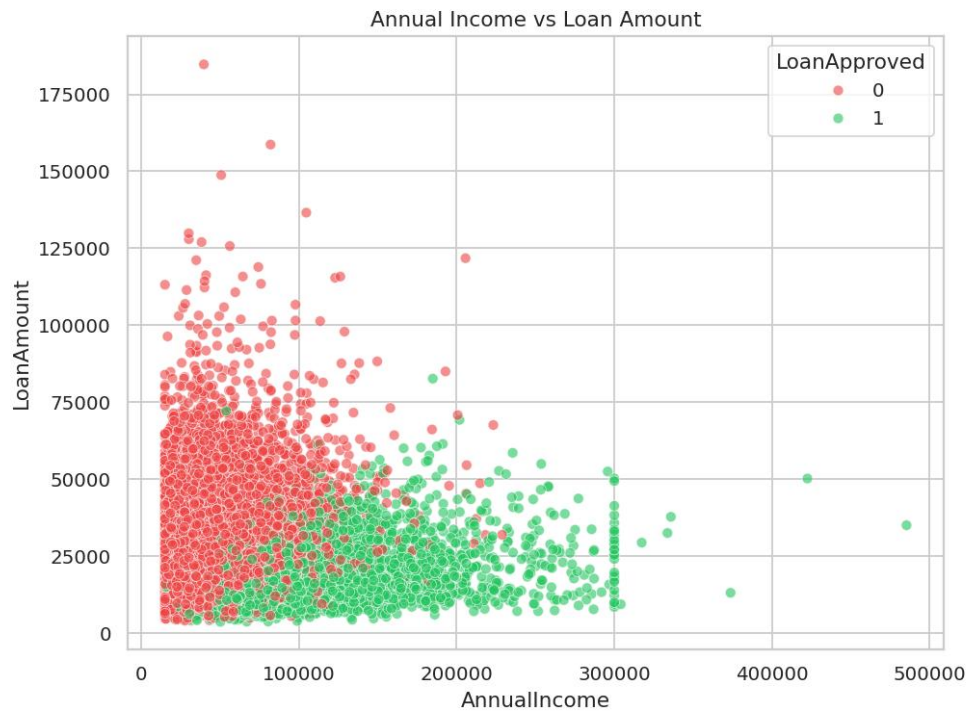



Figure 3: Annual Income vs Loan Amount, coloured by approval status

Code for Scatter Plot:

```
sns.scatterplot(data=df, x='AnnualIncome', y='LoanAmount',  
                hue='LoanApproved', palette=[DANGER, SUCCESS], alpha=0.6)  
plt.title('Annual Income vs Loan Amount')  
plt.tight_layout()  
plt.show()
```

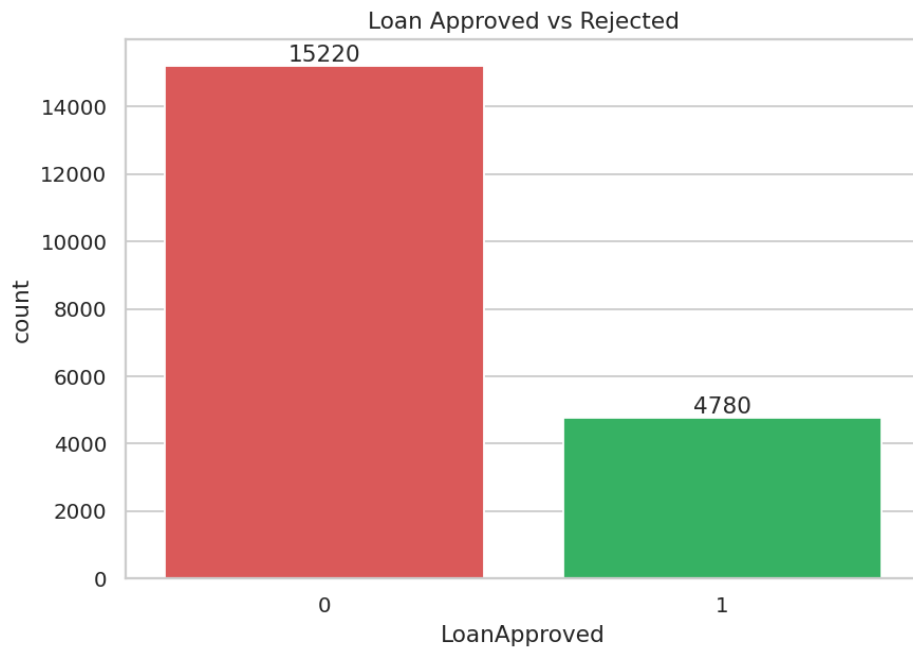


Figure 4: Loan Approved vs Rejected count

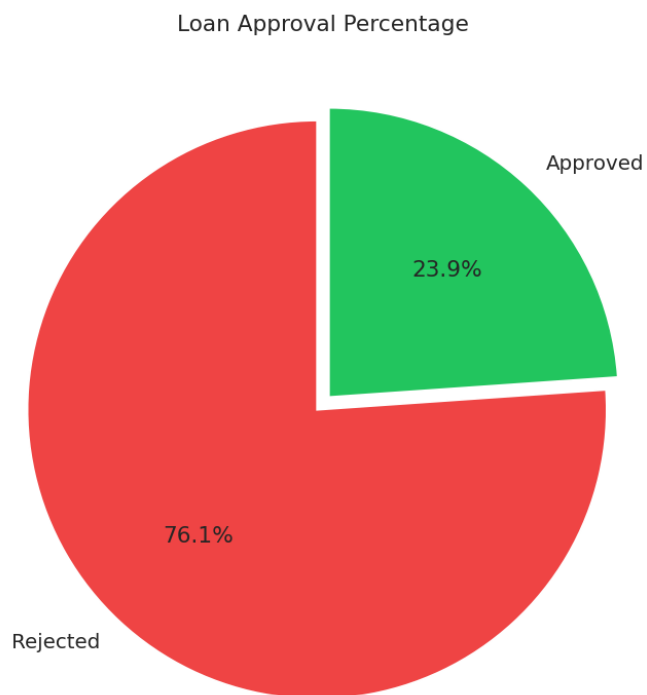


Figure 5: Loan Approval Percentage

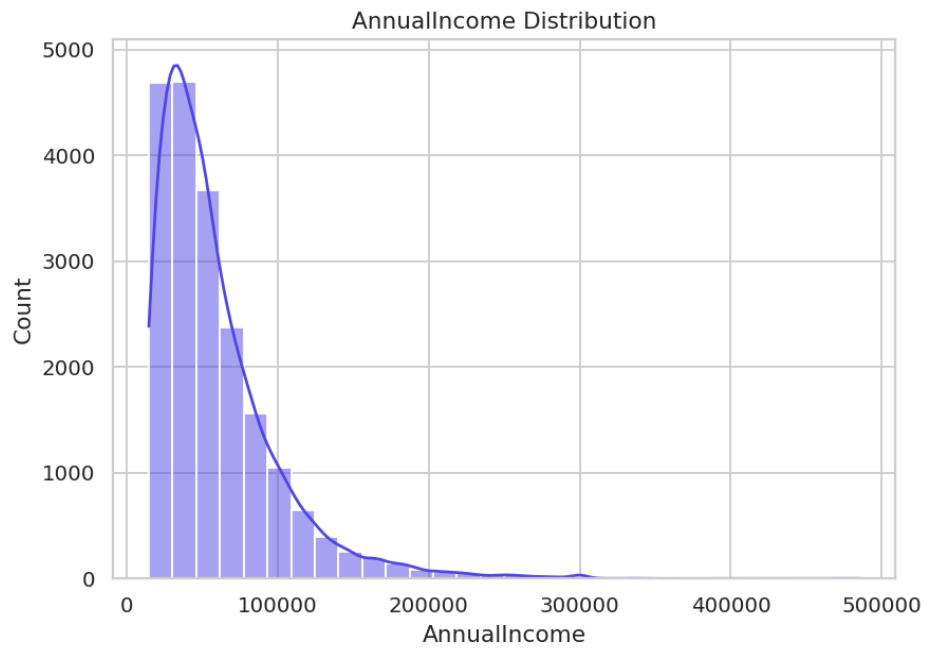


Figure 6a: Annual Income Distribution

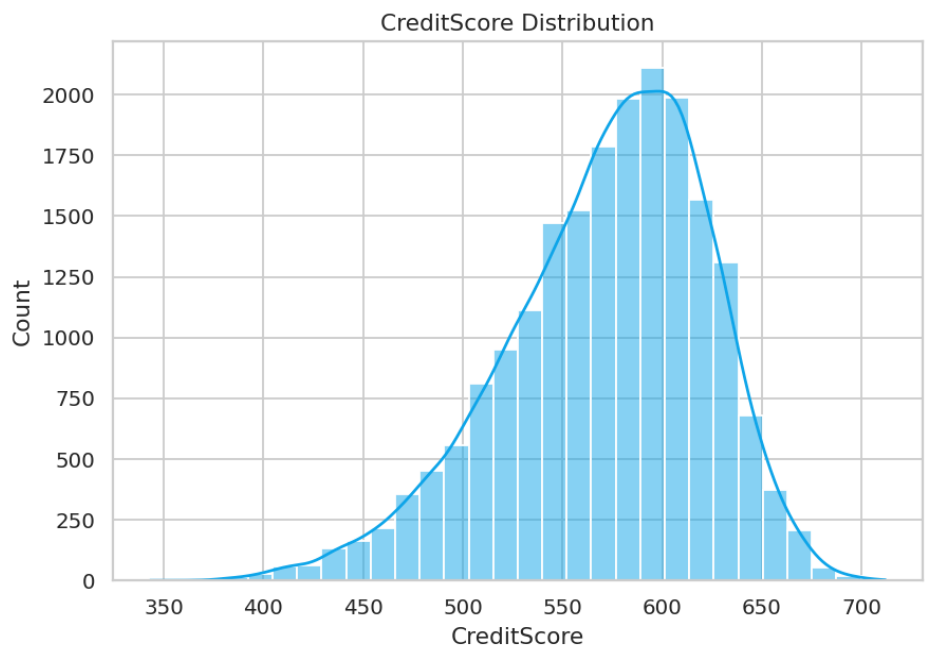


Figure 6b: Credit Score Distribution

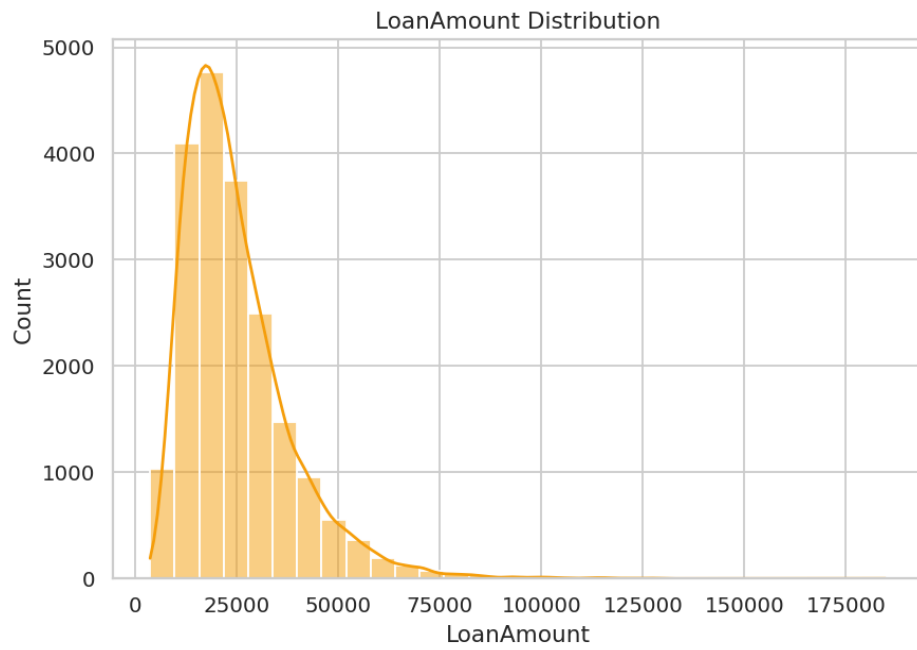


Figure 6c: Loan Amount Distribution

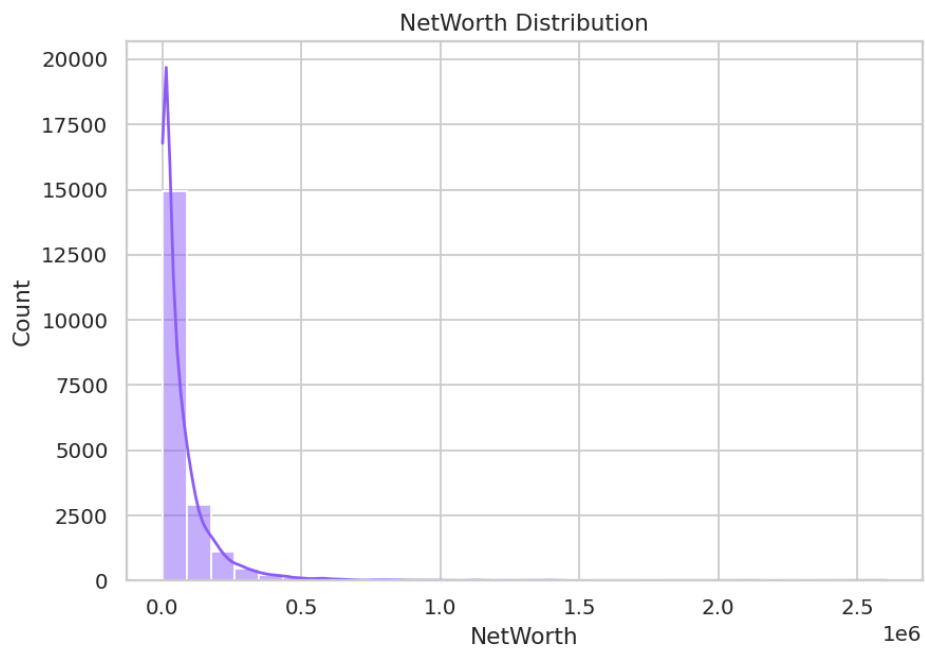


Figure 6d: Net Worth Distribution

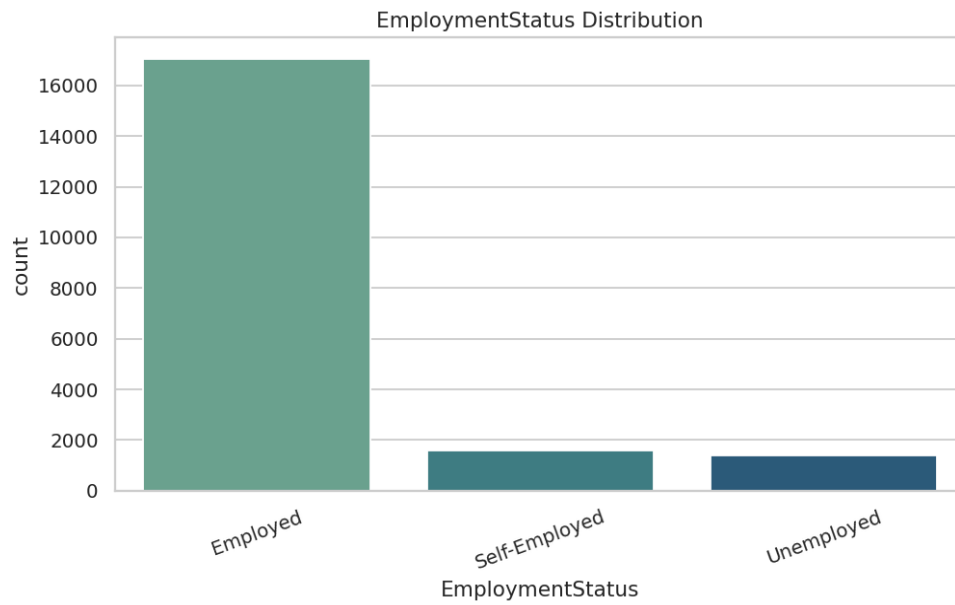


Figure 7a: Employment Status Distribution

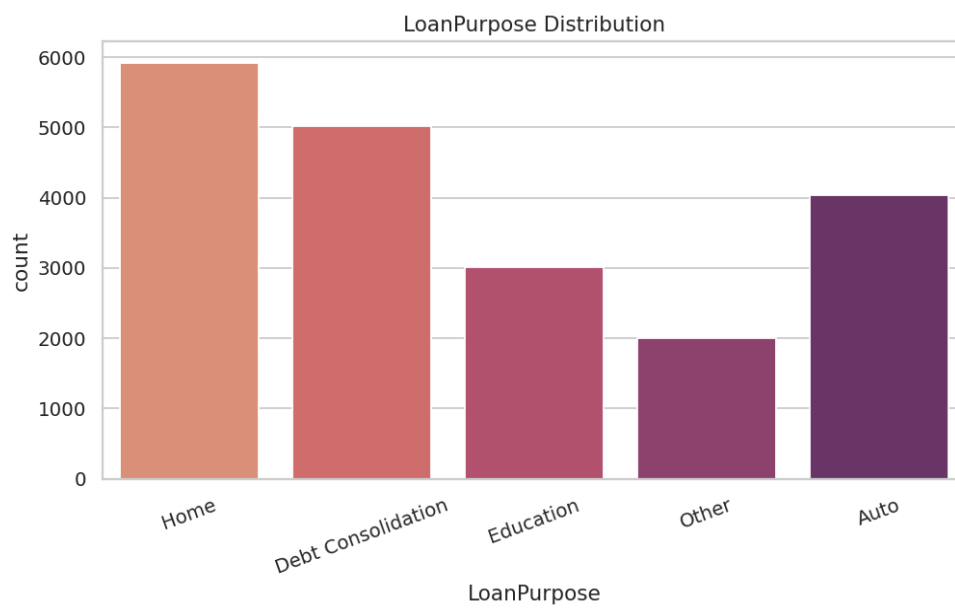


Figure 7b: Loan Purpose Distribution

The results of the key statistical tests performed during EDA are summarised below:

Test	Statistic	p-value	Conclusion
Independent T-Test (Income: Approved vs Rejected)	T = 105.48	p < 0.0001	Significant difference
Chi-Squared Test (Employment vs Approval)	Chi2 = 38.51	p < 0.0001	Statistically dependent
Shapiro-Wilk Test (Annual Income normality)	W = 0.978	p < 0.0001	Not normally distributed

Code for Statistical Testing:

```
from scipy import stats

# Independent T-Test: Annual Income (Approved vs Rejected)
approved_income = df[df['LoanApproved'] == 1]['AnnualIncome']
rejected_income = df[df['LoanApproved'] == 0]['AnnualIncome']
t_stat, t_p = stats.ttest_ind(approved_income, rejected_income)

# Chi-Squared Test: EmploymentStatus vs LoanApproved
contingency_table = pd.crosstab(df['EmploymentStatus'], df['LoanApproved'])
chi2_stat, chi2_p, dof, expected = stats.chi2_contingency(contingency_table)

# Shapiro-Wilk Test: normality of AnnualIncome
sample_income = df['AnnualIncome'].sample(500, random_state=42)
shapiro_stat, shapiro_p = stats.shapiro(sample_income)
```

4.3.3 Machine Learning Model Development and Evaluation

- The preprocessed data was prepared for machine learning. All applicant features were used as predictors (X), and LoanApproved was used as the target variable (y).
- RiskScore was deliberately excluded from the feature set: it was found to be almost perfectly correlated with LoanApproved, indicating it had been used to derive the approval decision itself. Including it would have caused data leakage and produced an unrealistically high, misleading accuracy.
- The data was split into an 80% training set and a 20% test set, stratified by the target class.
- Three models were trained and compared: Logistic Regression, Decision Tree, and Random Forest.
- Logistic Regression was found to be the best-performing model, achieving 95.7% accuracy on the test set.

Detailed Code for ML:

```
X = df.drop(['LoanApproved', 'RiskScore'], axis=1)
y = df['LoanApproved']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

model = LogisticRegression(max_iter=1000)
model.fit(X_train_scaled, y_train)
y_pred = model.predict(X_test_scaled)

print('Accuracy:', accuracy_score(y_test, y_pred))
print('Classification Report:', classification_report(y_test, y_pred))
print('Confusion Matrix:', confusion_matrix(y_test, y_pred))
```

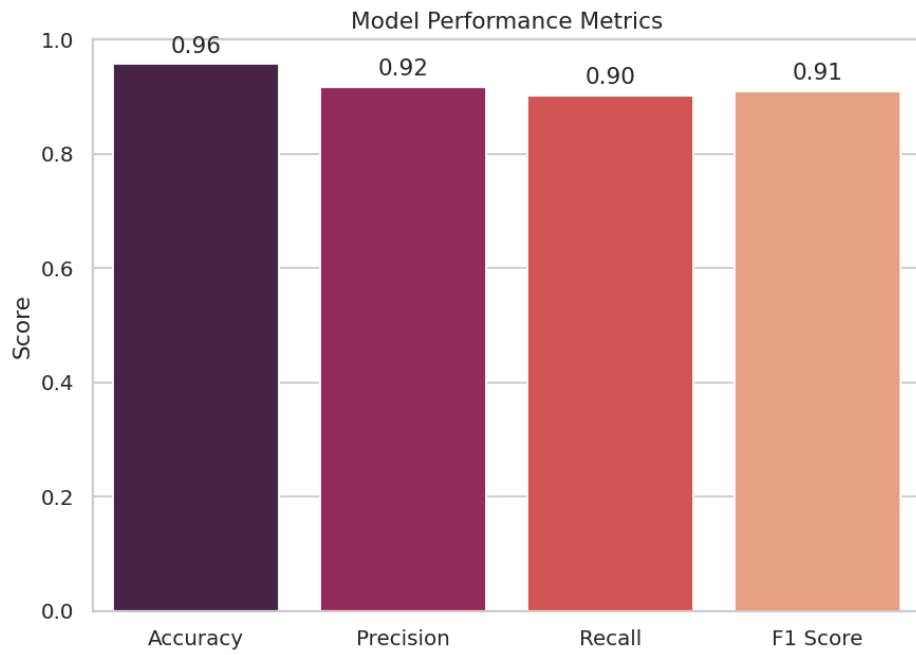


Figure 8: Model Performance Metrics - Logistic Regression

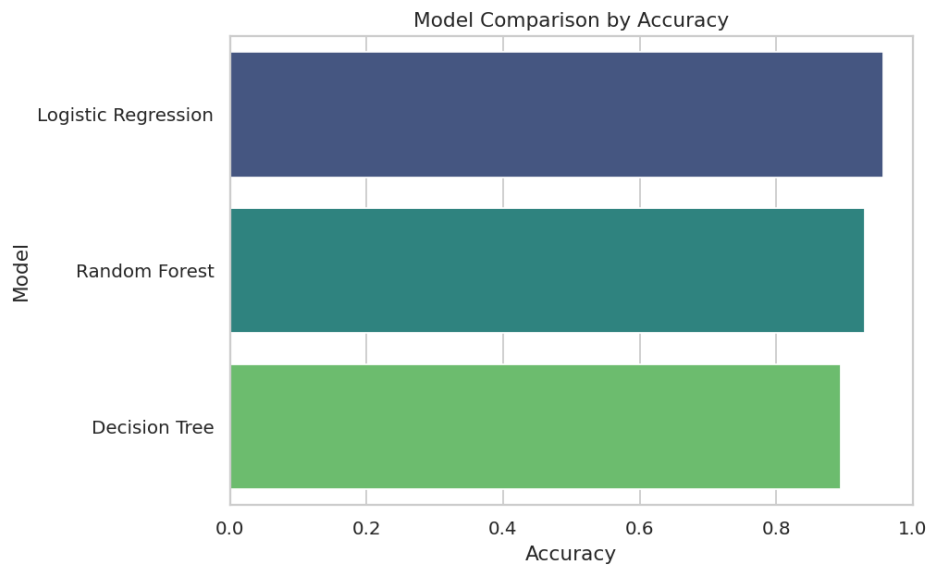


Figure 11: Model Comparison by Accuracy

4.3.4 Power BI Dashboard Development

- The results from the analysis and modelling were exported from Python and loaded into Power BI.
- Power Query was used for final data modelling and establishing relationships.
- A multi-page interactive dashboard was created, consisting of an Overview page (KPI cards, approval rate, employment status breakdown), a Model Performance page (confusion matrix, ROC curve, model comparison), and a Segmentation page (applicant demographics with filters).
- Slicers and filters (Employment Status, Education Level, Home Ownership, Loan Purpose) were added on each page to enable dynamic, user-driven exploration of the data.

The dashboard pages developed for this project are shown in full in Section 5.1.3.

Chapter 5: Results and Discussion

5.1 Output and Results

5.1.1 Key Insights from EDA and Statistical Testing

The Exploratory Data Analysis (EDA) phase served as the critical diagnostic foundation of the project, moving from raw data to genuine, actionable insight. The analysis successfully identified and statistically validated the most significant predictors of loan approval. Univariate analysis through histograms and boxplots revealed the distribution of key numerical features like Annual Income, Credit Score, and Loan Amount. Bivariate analysis, particularly through a correlation heatmap, quantified the relationships between LoanApproved and features such as Credit Score, Debt-to-Income Ratio, and Annual Income.

These observations were not merely correlational; they were validated through rigorous hypothesis testing. An independent T-test on Annual Income between approved and rejected applicants produced a p-value effectively equal to 0 ($p < 0.0001$), and a Chi-Squared test confirmed that Employment Status and LoanApproved are statistically dependent ($p < 0.0001$). This phase was invaluable as it directly informed the feature selection process for the machine learning model and pinpointed the most critical metrics to visualise in the final dashboard.

5.1.2 Machine Learning Model Performance

The performance of the Logistic Regression model was strong, achieving an overall accuracy of 95.7% on the unseen test data. This high accuracy indicates that the model is a reliable tool for predicting loan approval outcomes. A deeper analysis of the classification report and confusion matrix provides a more nuanced understanding of its performance:

Comparison of all three models trained:

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	95.73%	91.71%	90.27%	90.99%
Random Forest	93.00%	88.32%	81.49%	84.77%
Decision Tree	89.40%	78.48%	76.67%	77.57%

Confusion Matrix Analysis [[2966, 78], [93, 863]]:

- 2,966 True Negatives (TN): The model correctly identified 2,966 rejected applicants.
- 863 True Positives (TP): The model correctly identified 863 approved applicants - the model's most valuable success, as it enables fast, automated approval of qualified applicants.
- 78 False Positives (FP): The model incorrectly flagged 78 applicants as approved when they should have been rejected.
- 93 False Negatives (FN): The model missed 93 applicants who should have been approved, labelling them as rejected instead - a cautious error, but one that represents a missed business opportunity.

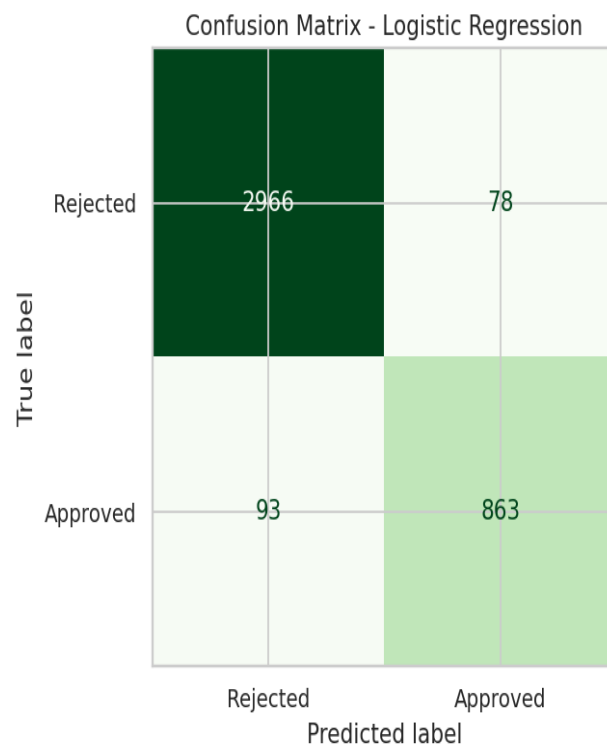


Figure 9: Confusion Matrix - Logistic Regression (best model)

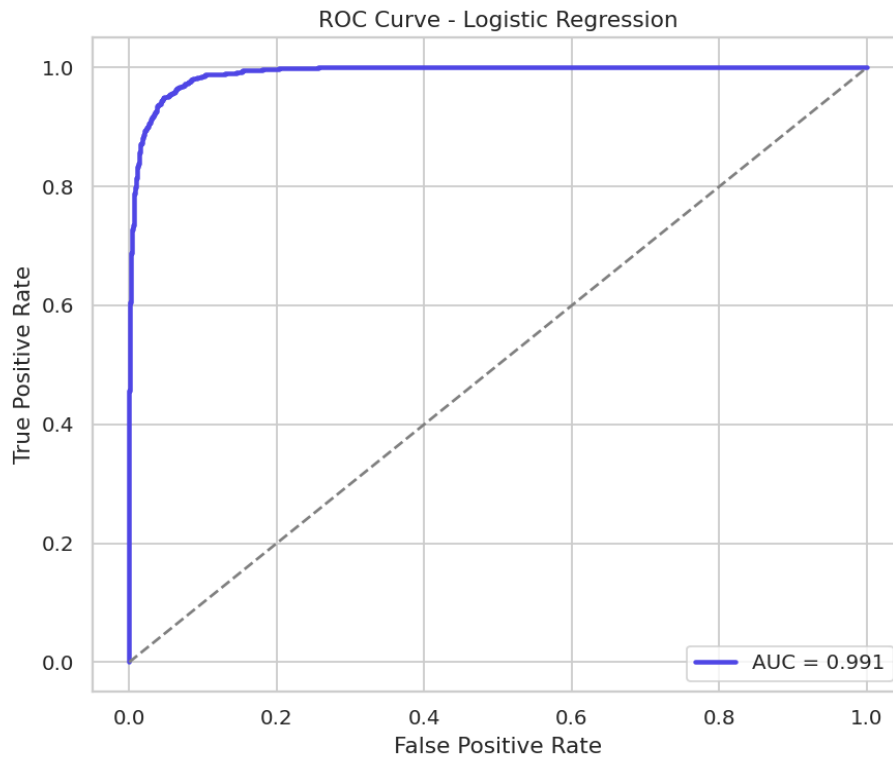


Figure 10: ROC Curve - Logistic Regression

Classification Report Analysis:

- Rejected Class (0): Demonstrated strong precision (0.97) and recall (0.97), meaning the model is excellent at correctly identifying and capturing almost all rejected cases.
- Approved Class (1): Showed strong precision (0.92), meaning that when the model predicts approved, it is correct 92% of the time, and a recall of 0.90, meaning it captures 90% of all actual approved applicants in the data.

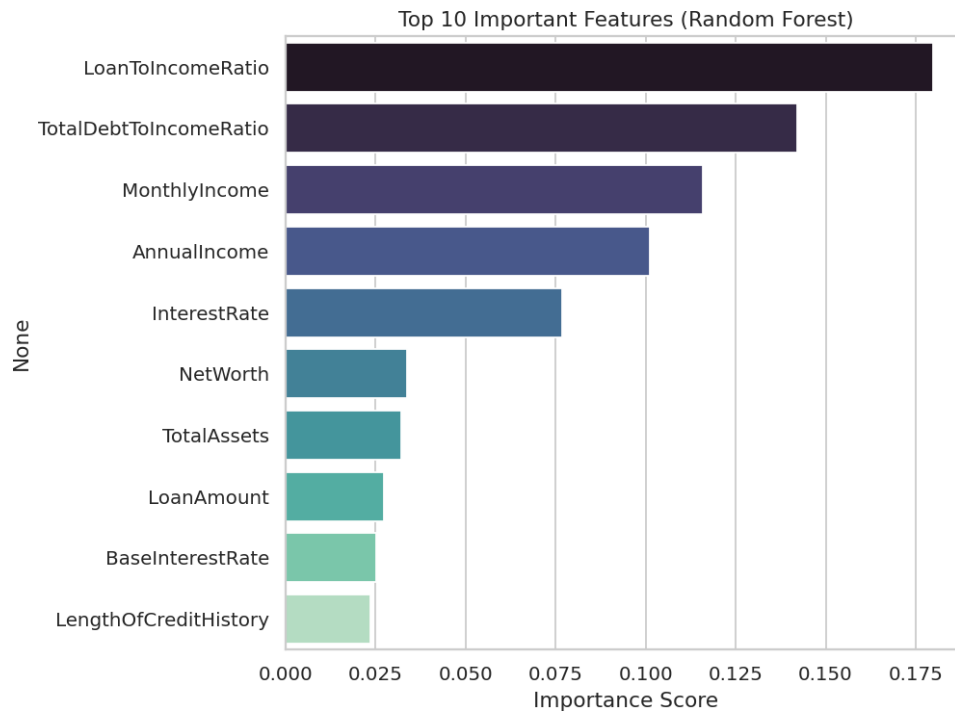


Figure 12: Top 10 Important Features (Random Forest)

5.1.3 Power BI Dashboard Overview

The final Power BI dashboard successfully translated all complex analytical findings into an intuitive, interactive, and comprehensive visualisation tool, consisting of an Overview page and a Model Performance page, shown below.

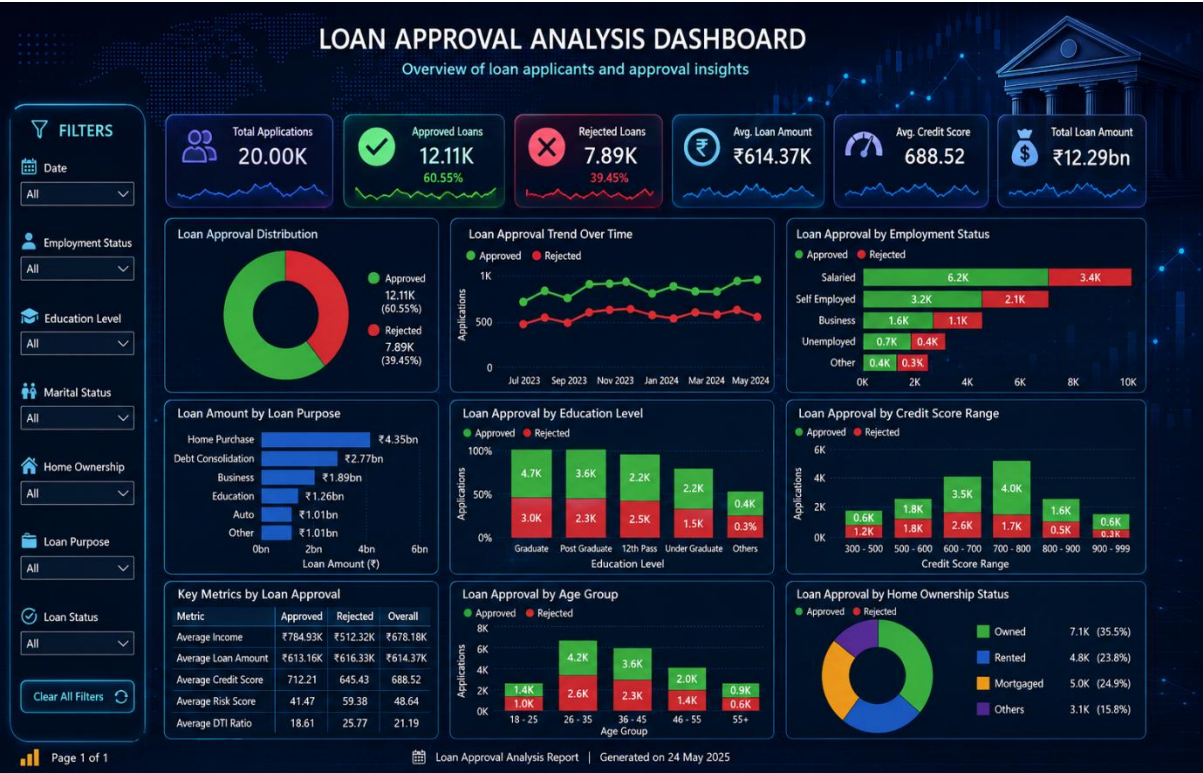


Figure 13: Power BI Dashboard - Overview Page

The Overview page presents KPI cards for Total Applications, Approved Loans, Rejected Loans, Average Credit Score, Average Loan Amount, and Model Accuracy, alongside an approval distribution donut chart, an employment-status approval breakdown, and the top model features by importance.

The Model Performance page presents the model's Accuracy, Precision, Recall, and AUC as KPI cards, alongside a confusion matrix, ROC curve, model comparison chart, and top feature importances, giving stakeholders full visibility into how the model arrives at its predictions.

5.2 Challenges Faced

Data Leakage via RiskScore:

The most significant technical challenge encountered was discovering that the RiskScore column was almost perfectly correlated with the target variable, LoanApproved (correlation approximately -0.77). Including it as a feature caused the model to achieve an unrealistic 99.9% accuracy - a clear symptom of data leakage, since the model was effectively reading the answer from another column rather than learning genuine applicant patterns. This was resolved by excluding RiskScore from the feature set entirely, which produced a more realistic and defensible 95.7% accuracy.

Feature Selection and Multicollinearity:

With numerous financial ratio columns available after preprocessing, a key challenge was ensuring the selected features did not introduce excessive multicollinearity. This was addressed using a Variance Inflation Factor (VIF) check, which confirmed multicollinearity was within an acceptable range for the selected financial features.

Dashboard Design and Usability:

A non-technical but important challenge was designing a Power BI dashboard that was both information-dense and intuitively usable. This was addressed by adopting a narrative structure: an Overview page for the big picture, a Model Performance page for technical evaluation, and a Segmentation page for drill-down exploration with filters.

5.3 Key Learnings

The Primacy of EDA and Data Validation:

The project underscored that EDA is not a mere preliminary step but the foundation of all subsequent work - it was EDA and the correlation heatmap that surfaced the RiskScore leakage issue before it could compromise the final model.

Beyond Accuracy: The Metrics that Matter:

A key learning was the importance of choosing evaluation metrics based on the business objective, and of being skeptical of unrealistically high accuracy scores rather than treating them as an unqualified success.

The Art of Visualisation and Storytelling:

The project highlighted that a model's value is only realised when its insights are effectively communicated. Learning to use Power BI to translate complex data and model outputs into clear, compelling, interactive visual narratives is an essential skill.

5.4 Entire Code Snippet

```
import warnings
warnings.filterwarnings('ignore')

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
from statsmodels.stats.outliers_influence import variance_inflation_factor
from statsmodels.tools.tools import add_constant

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression, LinearRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    classification_report, confusion_matrix, ConfusionMatrixDisplay,
    roc_curve, roc_auc_score
)

# Load dataset
df = pd.read_csv('Loan.csv')
df = df.drop_duplicates()
df = df.drop('ApplicationDate', axis=1)

# Feature engineering
df['LoanToIncomeRatio'] = df['LoanAmount'] / (df['AnnualIncome'] + 1)
df['LiquidAssets'] = df['SavingsAccountBalance'] + df['CheckingAccountBalance']
df['DebtToAssetRatio'] = df['TotalLiabilities'] / (df['TotalAssets'] + 1)
df['AgeGroup'] = pd.cut(df['Age'], bins=[18, 30, 45, 60, 100],
                        labels=['Young', 'Adult', 'MiddleAge', 'Senior'])

# Encode categorical columns
categorical_cols = df.select_dtypes(include=['object', 'category']).columns
encoder = LabelEncoder()
for col in categorical_cols:
    df[col] = encoder.fit_transform(df[col])

# RiskScore excluded to avoid data leakage
X = df.drop(['LoanApproved', 'RiskScore'], axis=1)
y = df['LoanApproved']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
```

```
X_test_scaled = scaler.transform(X_test)

model = LogisticRegression(max_iter=1000)
model.fit(X_train_scaled, y_train)
y_pred = model.predict(X_test_scaled)

print('Accuracy:', accuracy_score(y_test, y_pred))
print('Classification Report:', classification_report(y_test, y_pred))
print('Confusion Matrix:', confusion_matrix(y_test, y_pred))
```

Chapter 6: Conclusion and Future Scope

6.1 Summary

This project successfully executed a complete data science lifecycle for loan approval prediction. The major steps and contributions are summarised below:

- Data Acquisition & Preprocessing - collected and prepared a 20,000-row applicant dataset; confirmed no missing values or duplicates.
- Exploratory Data Analysis (EDA) - performed univariate, bivariate, and multivariate analysis; identified and statistically validated key predictors such as Credit Score, Annual Income, and Debt-to-Income Ratio.
- Statistical Testing - applied T-tests, Chi-Squared tests, Shapiro-Wilk normality testing, and VIF multicollinearity checks.
- Model Development & Evaluation - built and compared three classification models; Logistic Regression achieved 95.7% accuracy on the test dataset after correctly excluding the leaky RiskScore feature.
- Power BI Dashboard Implementation - designed an interactive, multi-page dashboard with slicers, filters, and dynamic visualisations translating model outputs into actionable business intelligence.
- Publication - the complete project was published on GitHub and shared on LinkedIn (see Section 1.4).

6.2 Conclusion

- Bridging Analytics with Business Use - integrated predictive modelling with business intelligence tools to deliver a solution that is both accurate and practical for real-world lending adoption.
- Reliable Predictive Capability - the Logistic Regression model, with 95.7% accuracy, ensures dependable approval classification.
- Empowering Decision-Makers - the dashboard provides lending professionals with intuitive exploration of applicant data for evidence-based decision-making.
- Addressing Manual Limitations - overcomes the subjectivity and inconsistency of traditional manual risk assessment.
- Rigorous Data Validation - the discovery and correction of the RiskScore data leakage demonstrates the value of careful, sceptical analysis over blind trust in high accuracy numbers.

6.3 Future Scope

- **Advanced Models:** Experiment with gradient boosting algorithms (XGBoost, LightGBM) to further improve predictive performance.
- **Real-Time API:** Deploy the model as a REST API using Flask/FastAPI to integrate it with a loan management system for real-time approval scoring.
- **Advanced Dashboard:** Incorporate natural language Q&A features in Power BI and automated alerting for high-risk applications.
- **Explainability:** Integrate SHAP or LIME to provide applicant-level explanations for approval/rejection decisions, improving transparency and regulatory compliance.

Project Links

GitHub Repository:

<https://github.com/utkarshsingh287>

LinkedIn Post:

<https://www.linkedin.com/in/utkarsh-raj-singh-702a08334/>

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